

OTT Platform Case Study Assignment – MongoDB Replication & Replica Sets

Case Study Title

Designing a Scalable OTT Streaming Analytics Platform using MongoDB Replica Sets

Collections and Columns

users

Column	Type	Description
userId	String	Unique user ID
name	String	User name
email	String	Email address
subscription	String	Basic/Premium
country	String	User country
createdAt	Date	Account creation date

streaming_history

Column	Type	Description
historyId	String	Watch history ID
userId	String	Linked user
contentTitle	String	Movie/Series name
genre	String	Content genre
watchDuration	Number	Minutes watched
watchedAt	Date	Watch timestamp

recommendations

Column	Type	Description
recommendationId	String	Recommendation ID
userId	String	Linked user
recommendedTitle	String	Suggested content
genre	String	Recommendation genre
score	Number	Recommendation score

Replica Set Configuration -

```
rs1 [direct: secondary] test> rs.status().members.forEach(m => {  
|   print(`${m.name} ---> ${m.stateStr}`);  
| })  
localhost:27017 ---> PRIMARY  
localhost:27018 ---> SECONDARY  
localhost:27019 ---> SECONDARY
```

CREATE

MONGODB-REPLICATION-OTT

data

db1

db2

db3

scripts

checkData.js

deleteData.js

failoverDemo.js

insertActivityLogs.js

insertRecommendations.js

insertStreamingHistory.js

insertUsers.js

setupReplica.js

updateData.js

OUTLINE

TIMELINE

scripts > JS setupReplica.js > members

```
1 rs.initiate({
2   _id:"rs1",
3   members:[
4     { _id:0,host:"localhost:27017"},
5     { _id:1,host:"localhost:27018"},
6     { _id:2,host:"localhost:27019"}
7   ]
8 });
9 print("Replica Set Initialized");
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\clouduser\seetha\mongodb-replication-ott> mongosh --port 27017
For mongosh info see: <https://www.mongodb.com/docs/mongodb-shell/>

The server generated these startup warnings when booting
2026-05-25T20:36:18.798+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
2026-05-25T20:36:18.799+05:30: This server is bound to localhost. Remote systems will be unable to connect to this server. Start the server with --bind_ip <address> to specify which IP addresses it should serve responses from, or with --bind_ip_all to bind to all interfaces. If this behavior is desired, start the server with --bind_ip 127.0.0.1 to disable this warning

rs1 [direct: primary] test> use ott
switched to db ott
rs1 [direct: primary] ott> show collections
recommendations
streaming_history
users
rs1 [direct: primary] ott>

READ

(using find() inside port 27017-primary)

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OUTLINE

TIMELINE

scripts > JS insertStreamingHistory.js > ...

```
1 use("ott");
2 db.streaming_history.insertMany([
3   {historyId:101,userId:1,contentTitle:"Stranger Things",genre:"Science-Fiction",watchDuration:360},
4   {historyId:102,userId:2,contentTitle:"Bridgerton",genre:"Period Drama",watchDuration:300},
5   {historyId:103,userId:3,contentTitle:"Dark",genre:"Science-Fiction",watchDuration:300}
6 ]);
7 print("Inserted Streaming History Successfully");
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\clouduser\seetha\mongodb-replication-ott> mongosh --port 27017
rs1 [direct: primary] ott> db.streaming_history.find()

[
 {
 _id: ObjectId('6a146683ea589705f3abc114'),
 historyId: 101,
 userId: 1,
 contentTitle: 'Stranger Things',
 genre: 'Science-Fiction',
 watchDuration: 360,
 watchedAt: '25-05-2026'
 },
 {
 _id: ObjectId('6a146683ea589705f3abc115'),
 historyId: 102,
 userId: 2,
 contentTitle: 'Bridgerton',
 genre: 'Period Drama',
 watchDuration: 300,
 watchedAt: '24-05-2026'
 },
 {

(using find() inside port 27018-secondary)

EXPLORER

MONGODB-REPLICATION-OTT

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updateData.js

OUTLINE

TIMELINE

JS insertUsers.js X JS insertStreamingHistory.js JS insertRecommendations.js JS insertActivityLogs.js

scripts > JS insertUsers.js > ...
1 use("ott");
2 db.users.insertMany([
3 {userid:1,name:"Anthony",email:"anthony@gmail.com",subscription:"basic",country:"UK",createdAt:"05-10-1999"},
4 {userid:2,name:"Eloise",email:"eloise@gmail.com",subscription:"premium",country:"USA",createdAt:"02-06-2005"},
5 {userid:3,name:"Kathany",email:"Kathany@gmail.com",subscription:"basic",country:"UK",createdAt:"05-10-1999"},
6]);
7 print("Users inserted Sucessfully");

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\clouduser\seetha\mongodb-replication-ott> mongosh --port 27018
rs1 [direct: secondary] ott> db.users.find()
[
 {
 _id: ObjectId('6a1466748317f686c8abc114'),
 userid: 1,
 name: 'Anthony',
 email: 'anthony@gmail.com',
 subscription: 'basic',
 country: 'UK',
 createdAt: '05-10-1999'
 },
 {
 _id: ObjectId('6a1466748317f686c8abc115'),
 userid: 2,
 name: 'Eloise',
 email: 'eloise@gmail.com',
 subscription: 'premium',
 country: 'USA',
 createdAt: '02-06-2005'
 }
]

Ln 7, Col 37 Spaces: 4 UTF-8 CRLF

(using find() inside port 27019-secondary)

MONGODB-REPLICATION-...

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OUTLINE

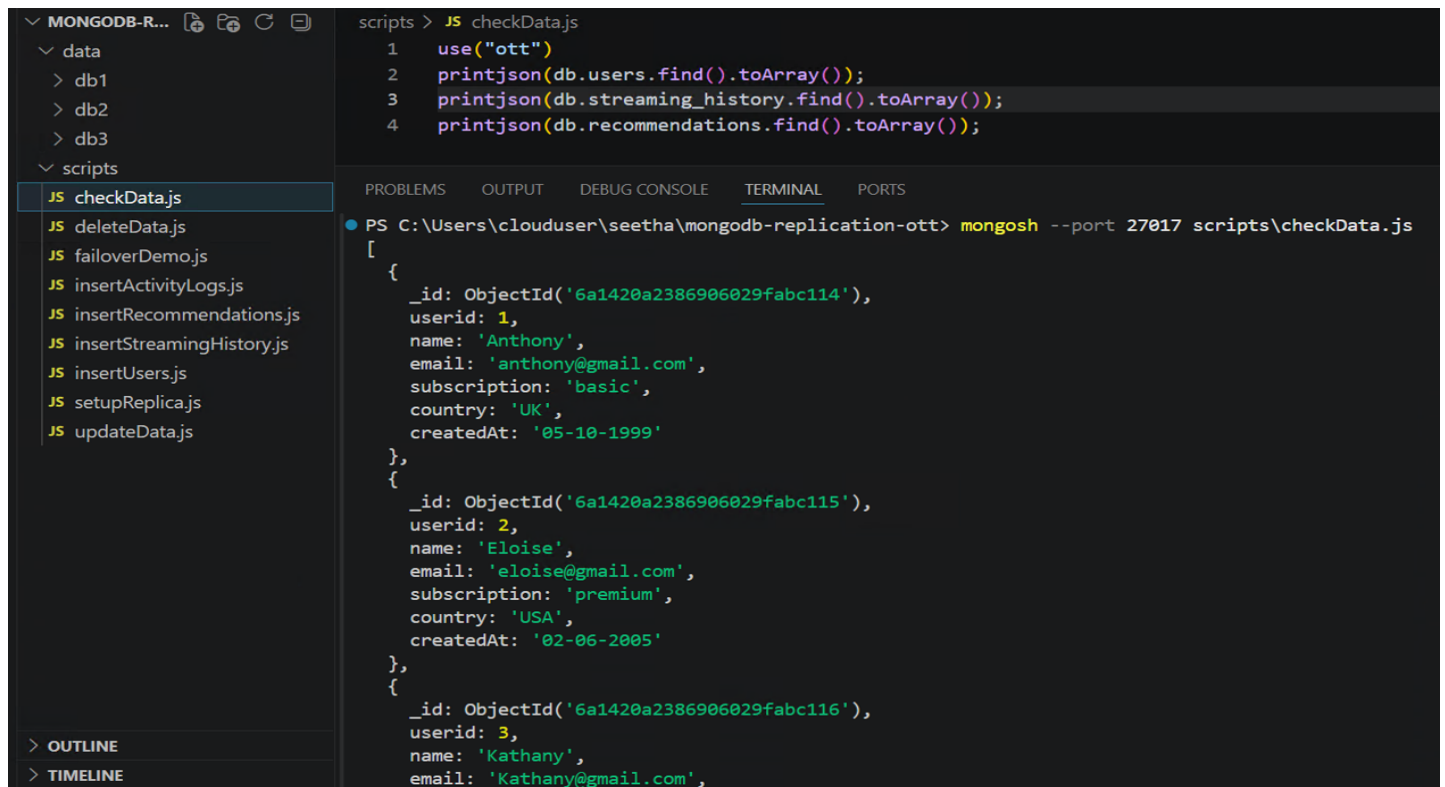
TIMELINE

scripts > JS insertRecommendations.js > ...
1 use("ott");
2 db.recommendations.insertMany([
3 {recommendationId:201,userId:1,recommendedTitle:"Avengers:Endgame",genre:"Action and Superhero",score:8},
4 {recommendationId:202,userId:2,recommendedTitle:"All of us are Dead",genre:"Thriller",score:7},
5 {recommendationId:203,userId:3,recommendedTitle:"Jumanji",genre:"Adventure",score:7},
6]);
7 print("Recommendations inserted sucessfully");

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\clouduser\seetha\mongodb-replication-ott> mongosh --port 27019
rs1 [direct: secondary] test> use ott
switched to db ott
rs1 [direct: secondary] ott> db.recommendations.find()
[
 {
 _id: ObjectId('6a1466a9d07cb45d3eabc116'),
 recommendationId: 203,
 userId: 3,
 recommendedTitle: 'Jumanji',
 genre: 'Adventure',
 score: 7
 },
 {
 _id: ObjectId('6a1466a9d07cb45d3eabc114'),
 recommendationId: 201,
 userId: 1,
 recommendedTitle: 'Avengers:Endgame',
 genre: 'Action and Superhero',
 score: 8
 }
]

Read from checkData.js

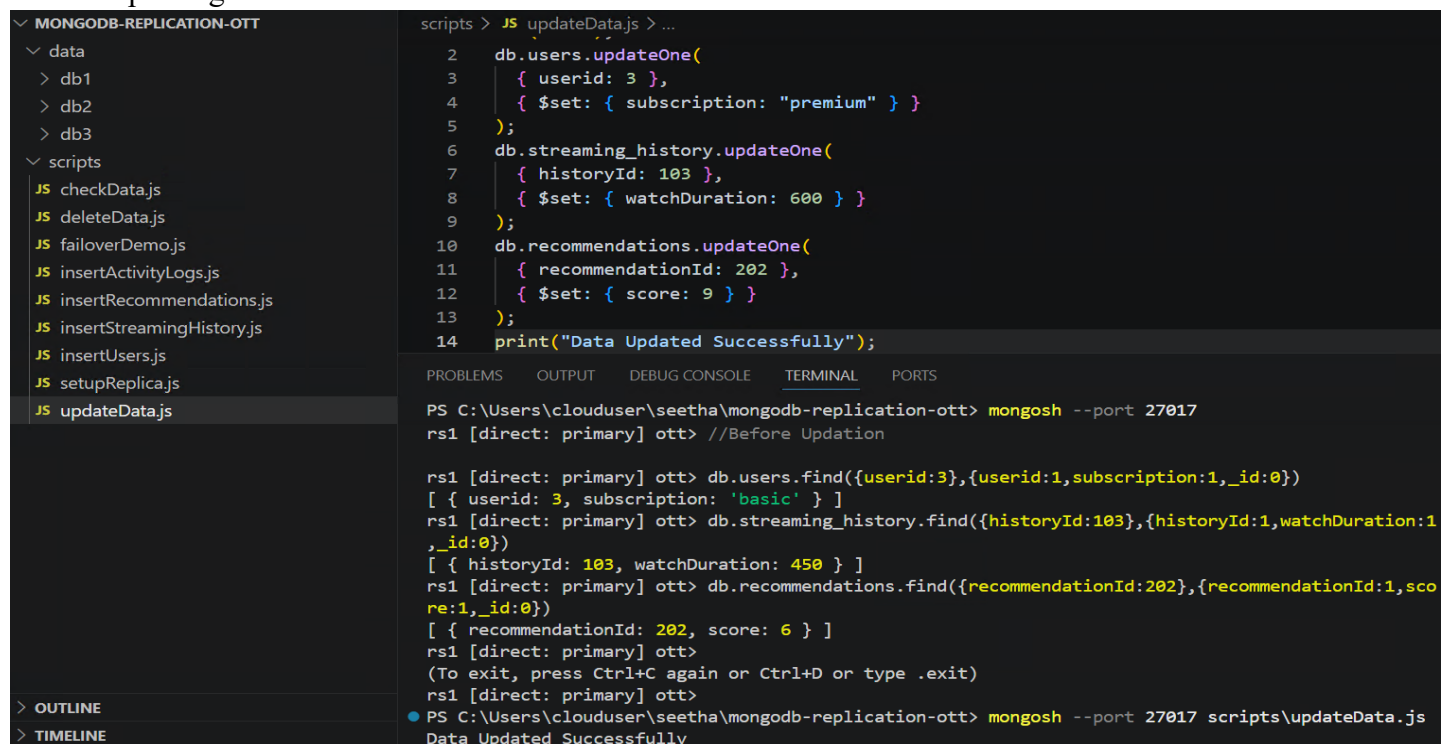


```
scripts > JS checkData.js
1 use("ott")
2 printjson(db.users.find().toArray());
3 printjson(db.streaming_history.find().toArray());
4 printjson(db.recommendations.find().toArray());
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Users\clouduser\seetha\mongodb-replication-ott> mongosh --port 27017 scripts\checkData.js
[
  {
    _id: ObjectId('6a1420a2386906029fab114'),
    userid: 1,
    name: 'Anthony',
    email: 'anthony@gmail.com',
    subscription: 'basic',
    country: 'UK',
    createdAt: '05-10-1999'
  },
  {
    _id: ObjectId('6a1420a2386906029fab115'),
    userid: 2,
    name: 'Eloise',
    email: 'eloise@gmail.com',
    subscription: 'premium',
    country: 'USA',
    createdAt: '02-06-2005'
  },
  {
    _id: ObjectId('6a1420a2386906029fab116'),
    userid: 3,
    name: 'Kathany',
    email: 'Kathany@gmail.com',
```

UPDATE

Before Updating



```
scripts > JS updateData.js > ...
2 db.users.updateOne(
3   { userid: 3 },
4   { $set: { subscription: "premium" } }
5 );
6 db.streaming_history.updateOne(
7   { historyId: 103 },
8   { $set: { watchDuration: 600 } }
9 );
10 db.recommendations.updateOne(
11   { recommendationId: 202 },
12   { $set: { score: 9 } }
13 );
14 print("Data Updated Successfully");
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\clouduser\seetha\mongodb-replication-ott> mongosh --port 27017
rs1 [direct: primary] ott> //Before Updation

rs1 [direct: primary] ott> db.users.find({userid:3},{userid:1,subscription:1,_id:0})
[ { userid: 3, subscription: 'basic' } ]
rs1 [direct: primary] ott> db.streaming_history.find({historyId:103},{historyId:1,watchDuration:1,_id:0})
[ { historyId: 103, watchDuration: 450 } ]
rs1 [direct: primary] ott> db.recommendations.find({recommendationId:202},{recommendationId:1,score:1,_id:0})
[ { recommendationId: 202, score: 6 } ]
rs1 [direct: primary] ott>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
rs1 [direct: primary] ott>
● PS C:\Users\clouduser\seetha\mongodb-replication-ott> mongosh --port 27017 scripts\updateData.js
Data Updated Successfully
```

After Updating

MONGODB-REPLICATION-OTT

data

db1

db2

db3

scripts

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setupReplica.js

updateData.js

OUTLINE

TIMELINE

scripts > JS updateData.js > ...

2 db.users.updateOne(
3 { userid: 3 },
4 { \$set: { subscription: "premium" } }
5);
6 db.streaming_history.updateOne(
7 { historyId: 103 },
8 { \$set: { watchDuration: 600 } }
9);
10 db.recommendations.updateOne(
11 { recommendationId: 202 },
12 { \$set: { score: 9 } }
13);
14 print("Data Updated Successfully");

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\clouduser\seetha\mongodb-replication-ott> mongosh --port 27017
rs1 [direct: primary] ott> use ott
switched to db ott
rs1 [direct: primary] ott> db.users.find({userid:3},{userid:1,subscription:1,_id:0})
[{ userid: 3, subscription: 'premium' }]
rs1 [direct: primary] ott> db.streaming_history.find({historyId:103},{historyId:1,watchDuration:1,_id:0})
[{ historyId: 103, watchDuration: 600 }]
rs1 [direct: primary] ott> db.recommendations.find({recommendationId:202},{recommendationId:1,score:1,_id:0})
[{ recommendationId: 202, score: 9 }]
rs1 [direct: primary] ott>

DELETE

MONGODB-REPLICATION-OTT

data

db1

db2

db3

scripts

checkData.js

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setupReplica.js

updateData.js

OUTLINE

TIMELINE

scripts > JS deleteData.js > ...

1 use("ott");
2 db.streaming_history.deleteOne({ historyId: 103 });
3 db.recommendations.deleteOne({ recommendationId: 202 });
4 db.users.deleteOne({ userid: 3 });
5 print("Data Deleted Successfully");

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\clouduser\seetha\mongodb-replication-ott> mongosh --port 27017
rs1 [direct: primary] ott> db.users.find({}, {userid:1,_id:0})
[{ userid: 1 }, { userid: 2 }]
rs1 [direct: primary] ott> db.streaming_history.find({}, {historyId:1,_id:0})
[{ historyId: 101 }, { historyId: 102 }]
rs1 [direct: primary] ott> db.recommendations.find({}, {recommendationId:1,_id:0})
[{ recommendationId: 201 }, { recommendationId: 203 }]
rs1 [direct: primary] ott>